Maximum Likelihood for a Gaussian Model

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Maximum likelihood for univariate Gaussian models

Suppose $X_1, \ldots, X_n \sim_{\text{iid}} \mathrm{N}(\mu, \sigma^2)$, and let $x_1, \ldots, x_n$ be the observed values. The parameter vector is $\underset{\sim}{\theta} = (\mu, \sigma^2)$. We treat σ^2, rather than σ, as the second parameter, and differentiate with respect to σ^2 directly.

By the likelihood of independent data, the likelihood is:

$$\mathcal{L}(\mu, \sigma^2) = \prod_{i=1}^{n} (2\pi\sigma^2)^{-1/2} \exp\left\{ -\frac{(x_i - \mu)^2}{2\sigma^2} \right\}$$

and by the log-likelihood of independent data, the log-likelihood is:

$$
\begin{aligned}
\ell(\mu, \sigma^2) &= \sum_{i=1}^{n} \log\left\{ (2\pi\sigma^2)^{-1/2} \exp\left\{ -\frac{(x_i - \mu)^2}{2\sigma^2} \right\} \right\} && \text{(log of each Gaussian density)} \\
&= \sum_{i=1}^{n} \left(\log\left\{ (2\pi\sigma^2)^{-1/2} \right\} + \log\left\{ \exp\left\{ -\frac{(x_i - \mu)^2}{2\sigma^2} \right\} \right\} \right) && \text{(log of a product)} \\
&= \sum_{i=1}^{n} \left(-\frac{1}{2} \log\left\{ 2\pi\sigma^2 \right\} + \log\left\{ \exp\left\{ -\frac{(x_i - \mu)^2}{2\sigma^2} \right\} \right\} \right) && \text{(log of a power)} \\
&= \sum_{i=1}^{n} \left(-\frac{1}{2} \log\left\{ 2\pi\sigma^2 \right\} - \frac{(x_i - \mu)^2}{2\sigma^2} \right) && \text{(log undoes exp)} \\
&= \sum_{i=1}^{n} \left(-\frac{1}{2} \log\left\{ 2\pi\sigma^2 \right\} \right) - \sum_{i=1}^{n} \frac{(x_i - \mu)^2}{2\sigma^2} && \text{(split the sum)} \\
&= -\frac{n}{2} \log\left\{ 2\pi\sigma^2 \right\} - \sum_{i=1}^{n} \frac{(x_i - \mu)^2}{2\sigma^2} && \text{(sum of } n \text{ identical terms)} \\
&= -\frac{n}{2} \log\left\{ 2\pi\sigma^2 \right\} - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 && \text{(factor the constant } \tfrac{1}{2\sigma^2} \text{ out of the sum)} \\
&= -\frac{n}{2} \left(\log\{2\pi\} + \log\{\sigma^2\} \right) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 && \text{(log of a product)} \\
&= -\frac{n}{2} \log\{2\pi\} - \frac{n}{2} \log\{\sigma^2\} - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 && \text{(distribute } -\tfrac{n}{2}\text{)}
\end{aligned}
$$

The score function

The score function is the vector of the two partial derivatives:

$$
\ell'(\mu, \sigma^2) = \begin{pmatrix} \frac{\partial}{\partial \mu} \ell(\mu, \sigma^2) \\ \frac{\partial}{\partial \sigma^2} \ell(\mu, \sigma^2) \end{pmatrix}
$$

For the first entry, only the last term of ℓ depends on μ:

$$
\begin{aligned}
\frac{\partial}{\partial\mu}\ell &= \frac{\partial}{\partial\mu}\left(-\frac{n}{2}\log\{2\pi\} - \frac{n}{2}\log\{\sigma^2\} - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right) && \text{(Gaussian log-likelihood)}\\
&= \frac{\partial}{\partial\mu}\Big(-\frac{n}{2}\log\{2\pi\}\Big) + \frac{\partial}{\partial\mu}\Big(-\frac{n}{2}\log\{\sigma^2\}\Big) + \frac{\partial}{\partial\mu}\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right) && \text{(linearity of differentiation)}\\
&= 0 + 0 + \frac{\partial}{\partial\mu}\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right) && \text{(derivative of a constant)}\\
&= \frac{\partial}{\partial\mu}\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right) && \text{(drop the zero terms)}\\
&= -\frac{1}{2\sigma^2}\frac{\partial}{\partial\mu}\sum_{i=1}^{n}(x_i-\mu)^2 && \text{(constant multiple rule)}\\
&= -\frac{1}{2\sigma^2}\sum_{i=1}^{n}\frac{\partial}{\partial\mu}(x_i-\mu)^2 && \text{(derivative of a sum is the sum of d}\\
&= -\frac{1}{2\sigma^2}\sum_{i=1}^{n}2(x_i-\mu)\frac{\partial}{\partial\mu}(x_i-\mu) && \text{(chain rule, outer function } u^2\text{)}
\end{aligned}
$$

The inner derivative is:

$$
\begin{aligned}
\frac{\partial}{\partial\mu}(x_i-\mu) &= \frac{\partial}{\partial\mu}x_i - \frac{\partial}{\partial\mu}\mu && \text{(linearity of differentiation)}\\
&= 0 - \frac{\partial}{\partial\mu}\mu && (x_i \text{ does not depend on } \mu)\\
&= 0 - 1 && \text{(derivative of } \mu \text{ with respect to itself)}\\
&= -1 && \text{(subtract)}
\end{aligned}
$$

Substituting the inner derivative back in:

$$
\begin{aligned}
\frac{\partial}{\partial \mu}\ell &= -\frac{1}{2\sigma^2}\sum_{i=1}^{n} 2(x_i-\mu)\cdot(-1) && \text{(inner derivative is } -1\text{)} \\
&= -\frac{1}{2\sigma^2}\sum_{i=1}^{n} -2(x_i-\mu) && \text{(multiply)} \\
&= -\frac{1}{2\sigma^2}\cdot(-2)\sum_{i=1}^{n}(x_i-\mu) && \text{(factor the constant } -2 \text{ out of the sum)} \\
&= \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) && \text{(multiply the constants)} \\
&= \frac{1}{\sigma^2}\left(\sum_{i=1}^{n} x_i - \sum_{i=1}^{n}\mu\right) && \text{(split the sum)} \\
&= \frac{1}{\sigma^2}\left(\sum_{i=1}^{n} x_i - n\mu\right) && \text{(sum of } n \text{ copies of } \mu\text{)}
\end{aligned}
$$

For the second entry, write the population variance σ^2 as a single variable v, so that the log-likelihood ℓ contains $-\frac{n}{2}\log v$ and $-\frac{1}{2}v^{-1}S$, where the sum of squared deviations $S = \sum_{i=1}^{n}\left(\left(\delta_\mu(x_i)\right)^2\right)$ from the mean parameter μ does not depend on σ^2:

$$
\begin{aligned}
\frac{\partial}{\partial \sigma^2}\ell &= \frac{\partial}{\partial \sigma^2}\left(-\frac{n}{2}\log\{2\pi\} - \frac{n}{2}\log\{\sigma^2\} - \frac{1}{2}(\sigma^2)^{-1}S\right) && \text{(Gaussian log-likelihood)} \\
&= \frac{\partial}{\partial \sigma^2}\left(-\frac{n}{2}\log\{2\pi\}\right) + \frac{\partial}{\partial \sigma^2}\left(-\frac{n}{2}\log\{\sigma^2\}\right) + \frac{\partial}{\partial \sigma^2}\left(-\frac{1}{2}(\sigma^2)^{-1}S\right) && \text{(linearity of differentiation)} \\
&= 0 + \frac{\partial}{\partial \sigma^2}\left(-\frac{n}{2}\log\{\sigma^2\}\right) + \frac{\partial}{\partial \sigma^2}\left(-\frac{1}{2}(\sigma^2)^{-1}S\right) && \text{(derivative of a constant)} \\
&= \frac{\partial}{\partial \sigma^2}\left(-\frac{n}{2}\log\{\sigma^2\}\right) + \frac{\partial}{\partial \sigma^2}\left(-\frac{1}{2}(\sigma^2)^{-1}S\right) && \text{(drop the zero term)} \\
&= \frac{\partial}{\partial \sigma^2}\left(-\frac{n}{2}\log\{\sigma^2\}\right) + \frac{\partial}{\partial \sigma^2}\left(-\frac{1}{2}S(\sigma^2)^{-1}\right) && \text{(reorder the factors)} \\
&= -\frac{n}{2}\frac{\partial}{\partial \sigma^2}\log\{\sigma^2\} - \frac{1}{2}S\frac{\partial}{\partial \sigma^2}(\sigma^2)^{-1} && \text{(constant multiple rule)} \\
&= -\frac{n}{2}(\sigma^2)^{-1} - \frac{1}{2}S\frac{\partial}{\partial \sigma^2}(\sigma^2)^{-1} && \text{(derivative of } \log v\text{)} \\
&= -\frac{n}{2}(\sigma^2)^{-1} - \frac{1}{2}S\left(-(\sigma^2)^{-2}\right) && \text{(power rule for } v^{-1}\text{)} \\
&= -\frac{n}{2}(\sigma^2)^{-1} + \frac{1}{2}S(\sigma^2)^{-2} && \text{(multiply)} \\
&= -\frac{n}{2}(\sigma^2)^{-1} + \frac{1}{2}(\sigma^2)^{-2}S && \text{(reorder the factors)}
\end{aligned}
$$

MLE of μ

Setting $\frac{\partial}{\partial \mu}\ell = 0$:

$$
\begin{aligned}
0 &= \frac{1}{\sigma^2}\left(\sum_{i=1}^{n} x_i - n\mu\right) && \text{(score for } \mu\text{)} \\
0 &= \sum_{i=1}^{n} x_i - n\mu && \text{(multiply both sides by } \sigma^2\text{)} \\
n\mu &= \sum_{i=1}^{n} x_i && \text{(add } n\mu \text{ to both sides)} \\
\mu &= \frac{1}{n}\sum_{i=1}^{n} x_i && \text{(divide by } n\text{)} \\
\mu &= \bar{x} && \text{(definition of } \bar{x}\text{)}
\end{aligned}
$$

This solution does not depend on σ^2. The second derivative is

$$
\begin{aligned}
\frac{\partial^2 \ell}{\partial \mu^2} &= \frac{\partial}{\partial \mu}\left(\frac{1}{\sigma^2}\left(\sum_{i=1}^{n} x_i - n\mu\right)\right) && \text{(differentiate the score for } \mu\text{)} \\
&= \frac{1}{\sigma^2}\frac{\partial}{\partial \mu}\left(\sum_{i=1}^{n} x_i - n\mu\right) && \text{(constant multiple rule)} \\
&= \frac{1}{\sigma^2}\left(\frac{\partial}{\partial \mu}\sum_{i=1}^{n} x_i - \frac{\partial}{\partial \mu}(n\mu)\right) && \text{(linearity of differentiation)} \\
&= \frac{1}{\sigma^2}\left(0 - \frac{\partial}{\partial \mu}(n\mu)\right) && (\textstyle\sum_{i=1}^{n} x_i \text{ does not depend on } \mu) \\
&= \frac{1}{\sigma^2}\left(0 - n\frac{\partial}{\partial \mu}\mu\right) && \text{(constant multiple rule)} \\
&= \frac{1}{\sigma^2}(0 - n) && \text{(derivative of } \mu \text{ with respect to itself)} \\
&= \frac{1}{\sigma^2}(-n) && \text{(subtract)} \\
&= -\frac{n}{\sigma^2} && \text{(multiply)} \\
&< 0, && (n > 0 \text{ and } \sigma^2 > 0)
\end{aligned}
$$

so for every fixed σ^2, ℓ is maximized over μ at $\bar{x}$, and $\hat{\mu}_{\mathrm{ML}} = \bar{x}$.

MLE of σ^2

Setting $\frac{\partial}{\partial \sigma^2}\ell = 0$:

$$\begin{aligned}
0 &= -\frac{n}{2}(\sigma^2)^{-1} + \frac{1}{2}(\sigma^2)^{-2}\sum_{i=1}^{n}(x_i-\mu)^2 && \text{(score for } \sigma^2\text{)} \\
\frac{n}{2}(\sigma^2)^{-1} &= \frac{1}{2}(\sigma^2)^{-2}\sum_{i=1}^{n}(x_i-\mu)^2 && \text{(add } \tfrac{n}{2}(\sigma^2)^{-1}\text{)} \\
n\sigma^2 &= \sum_{i=1}^{n}(x_i-\mu)^2 && \text{(multiply both sides by } 2(\sigma^2)^2\text{)} \\
\sigma^2 &= \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2 && \text{(divide both sides by } n\text{)}
\end{aligned}$$

Substituting the maximizer $\mu = \bar{x}$, which does not depend on σ^2, maximizes the profile log-likelihood of the Gaussian example, and gives:

$$\hat{\sigma}^2_{\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2$$

The profile log-likelihood, $\ell_p(\sigma^2) = -\frac{n}{2}\log\{2\pi\} - \frac{n}{2}\log\{\sigma^2\} - \frac{n\hat{\sigma}^2_{\mathrm{ML}}}{2\sigma^2}$, increases for $\sigma^2 < \hat{\sigma}^2_{\mathrm{ML}}$ and decreases for $\sigma^2 > \hat{\sigma}^2_{\mathrm{ML}}$, because its derivative, $\frac{n}{2}(\sigma^2)^{-2}(\hat{\sigma}^2_{\mathrm{ML}} - \sigma^2)$, has the sign of $\hat{\sigma}^2_{\mathrm{ML}} - \sigma^2$. So $(\bar{x}, \hat{\sigma}^2_{\mathrm{ML}})$ is the global maximizer, provided the x_i are not all equal.

> **Remark (Differentiating with respect to the variance or the precision)**
>
> *Remark* 0.1 (Differentiating with respect to the variance or the precision). Differentiating with respect to σ^2 as a single variable, rather than with respect to σ, keeps the algebra short. Replacing σ^2 with the precision $\tau \stackrel{\text{def}}{=} 1/\sigma^2$ and differentiating with respect to τ can be shorter still, because τ enters the log-likelihood as $\frac{n}{2}\log\tau - \frac{\tau}{2}\sum_{i=1}^{n}(x_i-\mu)^2$. By the invariance of maximum likelihood estimates, $\hat{\tau}_{\mathrm{ML}} = 1/\hat{\sigma}^2_{\mathrm{ML}}$.

> **Remark (The Gaussian variance MLE is biased)**
>
> *Remark* 0.2 (The Gaussian variance MLE is biased). This MLE divides by n, so it is a biased estimator of σ^2 (bias of the divide-by-n estimator).

Second derivatives

The remaining second derivatives are:

$$
\begin{aligned}
\frac{\partial^2 \ell}{\partial (\sigma^2)^2} &= \frac{\partial}{\partial \sigma^2}\left(-\frac{n}{2}(\sigma^2)^{-1} + \frac{1}{2}(\sigma^2)^{-2}\sum_{i=1}^{n}(x_i-\mu)^2 \right) && \text{(differentiate the score for } \sigma^2\text{)} \\
&= \frac{\partial}{\partial \sigma^2}\Big(-\frac{n}{2}(\sigma^2)^{-1} \Big) + \frac{\partial}{\partial \sigma^2}\left(\frac{1}{2}(\sigma^2)^{-2}\sum_{i=1}^{n}(x_i-\mu)^2 \right) && \text{(linearity of differentiation)} \\
&= \frac{\partial}{\partial \sigma^2}\Big(-\frac{n}{2}(\sigma^2)^{-1} \Big) + \frac{\partial}{\partial \sigma^2}\left(\frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2(\sigma^2)^{-2} \right) && \text{(reorder the factors)} \\
&= -\frac{n}{2}\frac{\partial}{\partial \sigma^2}(\sigma^2)^{-1} + \frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2\frac{\partial}{\partial \sigma^2}(\sigma^2)^{-2} && \text{(constant multiple rule)} \\
&= -\frac{n}{2}(-(\sigma^2)^{-2}) + \frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2\frac{\partial}{\partial \sigma^2}(\sigma^2)^{-2} && \text{(power rule for } v^{-1}\text{)} \\
&= -\frac{n}{2}(-(\sigma^2)^{-2}) + \frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2(-2(\sigma^2)^{-3}) && \text{(power rule for } v^{-2}\text{)} \\
&= \frac{n}{2}(\sigma^2)^{-2} - \sum_{i=1}^{n}(x_i-\mu)^2(\sigma^2)^{-3} && \text{(multiply the constants)} \\
&= \frac{n}{2}(\sigma^2)^{-2} - (\sigma^2)^{-3}\sum_{i=1}^{n}(x_i-\mu)^2 && \text{(reorder the factors)} \\
\frac{\partial^2 \ell}{\partial \mu\, \partial \sigma^2} &= \frac{\partial}{\partial \mu}\left(-\frac{n}{2}(\sigma^2)^{-1} + \frac{1}{2}(\sigma^2)^{-2}\sum_{i=1}^{n}(x_i-\mu)^2 \right) && \text{(differentiate the score for } \sigma^2\text{)} \\
&= \frac{\partial}{\partial \mu}\Big(-\frac{n}{2}(\sigma^2)^{-1} \Big) + \frac{\partial}{\partial \mu}\left(\frac{1}{2}(\sigma^2)^{-2}\sum_{i=1}^{n}(x_i-\mu)^2 \right) && \text{(linearity of differentiation)} \\
&= 0 + \frac{\partial}{\partial \mu}\left(\frac{1}{2}(\sigma^2)^{-2}\sum_{i=1}^{n}(x_i-\mu)^2 \right) && \text{(derivative of a constant)} \\
&= \frac{\partial}{\partial \mu}\left(\frac{1}{2}(\sigma^2)^{-2}\sum_{i=1}^{n}(x_i-\mu)^2 \right) && \text{(drop the zero term)} \\
&= \frac{1}{2}(\sigma^2)^{-2}\frac{\partial}{\partial \mu}\sum_{i=1}^{n}(x_i-\mu)^2 && \text{(constant multiple rule)} \\
&= \frac{1}{2}(\sigma^2)^{-2}\sum_{i=1}^{n}\frac{\partial}{\partial \mu}(x_i-\mu)^2 && \text{(derivative of a sum is the sum of derivatives)} \\
&= \frac{1}{2}(\sigma^2)^{-2}\sum_{i=1}^{n}-2(x_i-\mu) && \text{(chain-rule result from the score for } \mu\text{)} \\
&= \frac{1}{2}(\sigma^2)^{-2}\cdot(-2)\sum_{i=1}^{n}(x_i-\mu) && \text{(factor the constant } -2 \text{ out of the sum)} \\
&= -(\sigma^2)^{-2}\sum_{i=1}^{n}(x_i-\mu) && \text{(multiply the constants)}
\end{aligned}
$$

At the MLE, $\sum_{i=1}^n (x_i - \bar{x}) = 0$ and $\sum_{i=1}^n (x_i - \bar{x})^2 = n\hat{\sigma}^2$ (writing $\hat{\sigma}^2$ for $\hat{\sigma}^2_{\mathrm{ML}}$), so:

$$\begin{aligned}
\left.\frac{\partial^2 \ell}{\partial (\sigma^2)^2}\right|_{\mathrm{MLE}} &= \frac{n}{2}(\hat{\sigma}^2)^{-2} - (\hat{\sigma}^2)^{-3}\sum_{i=1}^n (x_i - \bar{x})^2 && \text{(evaluate at } \mu = \bar{x},\ \sigma^2 = \hat{\sigma}^2\text{)} \\
&= \frac{n}{2}(\hat{\sigma}^2)^{-2} - (\hat{\sigma}^2)^{-3} n\hat{\sigma}^2 && \text{(substitute } \textstyle\sum_{i=1}^n (x_i - \bar{x})^2 = n\hat{\sigma}^2\text{)} \\
&= \frac{n}{2}(\hat{\sigma}^2)^{-2} - n(\hat{\sigma}^2)^{-3}\hat{\sigma}^2 && \text{(reorder the factors)} \\
&= \frac{n}{2}(\hat{\sigma}^2)^{-2} - n(\hat{\sigma}^2)^{-2} && \text{(cancel one factor of } \hat{\sigma}^2\text{)} \\
&= -\frac{n}{2}(\hat{\sigma}^2)^{-2} && \text{(combine like terms)} \\
\left.\frac{\partial^2 \ell}{\partial \mu\, \partial \sigma^2}\right|_{\mathrm{MLE}} &= -(\hat{\sigma}^2)^{-2}\sum_{i=1}^n (x_i - \bar{x}) && \text{(evaluate at } \mu = \bar{x},\ \sigma^2 = \hat{\sigma}^2\text{)} \\
&= -(\hat{\sigma}^2)^{-2} \cdot 0 && \text{(substitute } \textstyle\sum_{i=1}^n (x_i - \bar{x}) = 0\text{)} \\
&= 0 && \text{(multiply by zero)}
\end{aligned}$$

Information matrix and standard errors

Collecting the second derivatives at the MLE, the observed information is

$$I(\hat{\mu}, \hat{\sigma}^2) = \begin{bmatrix} \frac{n}{\hat{\sigma}^2} & 0 \\ 0 & \frac{n}{2(\hat{\sigma}^2)^2} \end{bmatrix}$$

Its diagonal entries are positive and its off-diagonal entries are zero, so it is positive definite, consistent with the MLE being a maximum. The inverse of a diagonal matrix inverts each diagonal entry, so:

$$\left(I(\hat{\mu}, \hat{\sigma}^2)\right)^{-1} = \begin{bmatrix} \frac{\hat{\sigma}^2}{n} & 0 \\ 0 & \frac{2(\hat{\sigma}^2)^2}{n} \end{bmatrix}$$

By the asymptotic distribution of the MLE, the estimated standard errors are $\widehat{\mathrm{SE}}(\hat{\mu}) = \hat{\sigma}/\sqrt{n}$ and $\widehat{\mathrm{SE}}(\hat{\sigma}^2) = \hat{\sigma}^2\sqrt{2/n}$, and the two estimates are approximately uncorrelated in large samples.

See also [1], Example 7.2.12.

[1] G. Casella and R. Berger, *Statistical inference*, 2nd ed. Cengage Learning, 2002. Available: https://www.cengage.com/c/statistical-inference-2e-casella-berger/9780534243128/